

Amogh Merudi

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EXPERIENCE

Full-Stack Developer Intern

GrowME AI — AI-Powered College Readiness Platform

Apr 2026 – Present

- Building core modules of a production web application using Next.js 14 (App Router) and Tailwind CSS on the frontend, with a FastAPI/Python backend serving a PostgreSQL database on AWS RDS.
- Implementing role-based authentication and session management (Student, Counselor, Admin) with JWT tokens and Auth0, enforcing secure access control across the full application.
- Contributing to a CI/CD pipeline via GitHub Actions for automated testing and deployment across separate staging and production environments hosted on AWS ECS.
- Architected a multi-model AI scoring pipeline combining XGBoost admit probability inference, a rules-based fit scorer, and a Claude API qualitative gap analyzer to generate per-university admission scores for students across 45 US institutions.
- Built the feature engineering pipeline that normalizes student DNA vectors (GPA, SAT, AP count, extracurricular count, intended major, dimension scores) for XGBoost inference, with synthetic training labels derived from Common Data Set admission statistics.
- Designed and implemented async Celery rescore workers with lock contention handling, supersede logic, and monotone floor gating to ensure score consistency and prevent stale results from surfacing in the UI.
- Integrated CDS-verified admission criteria from 45 universities into the scoring engine, powering per-school rubric-based fit scoring via a criteria-driven scoring helper consumed by the rescore coordinator.

Co-Founder & CTO

Arclo — AI Growth OS for Early-Stage SaaS

Jan 2026 – Present

- Architected and shipped a full-stack product from scratch using a React/TypeScript frontend and FastAPI/Python backend, iterating rapidly across the entire stack in a fast-paced, resource-constrained environment.
- Built and maintained two production web interfaces for AI agents (Patch and Flux), implementing component-driven UI patterns, responsive layouts, and clean state management across the React frontend.
- Designed RESTful API contracts between frontend and backend, coordinating data flow for real-time agent outputs including friction detection reports and growth experiment recommendations.
- Established frontend development conventions and a versioned codebase on GitHub, enabling consistent iterative delivery and reproducible builds across feature branches.

R&D Intern

Tree House Ventures

Oct 2025 – Jan 2026

- Engineered a full-stack AI price scraping platform using Python/FastAPI for the backend and a web-based monitoring interface, delivering structured data extraction from unstructured web content with near real-time updates.
- Collaborated cross-functionally in a remote environment to evaluate a retrieval-augmented generation framework, documenting findings and translating technical results into structured reports for the broader team.

Market Expansion & Research Associate

GrowME AI — AI-Powered College Readiness Platform

Apr 2025 – Sept 2025

- Researched and structured domain knowledge for integration into an LLM-based student advisory product, translating complex admissions systems into clean, structured onboarding content consumed by the application layer.

Director of Technology

Ontario Youth Sustainability Solution (OYSS)

Mar 2024 – Sept 2025

- Led end-to-end development of the official OYSS platform in React, TypeScript, and Tailwind CSS, delivering a modular content system for events, blogs, and podcasts serving a national nonprofit audience.
- Implemented responsive, accessibility-aware component patterns and dynamic content rendering, reducing manual update overhead for non-technical team members across the organization.

PROJECTS

Pitwall — F1 Pit Strategy Optimizer & Race Analytics Dashboard | *FastAPI · Next.js · TypeScript · Vercel · Railway*

- Built a full-stack F1 strategy platform with a Next.js/TypeScript frontend and FastAPI backend, featuring live mode, head-to-head comparison, what-if simulator, and sector-level timing analysis.
- Designed and implemented a geospatial data dashboard with complex real-time data rendering, deploying the backend on Railway and the frontend on Vercel with continuous delivery from GitHub.
- Engineered tyre degradation modelling, crossover lap calculations, and undercut threat scoring across 2018+ race sessions, serving structured strategy output to the frontend via clean REST endpoints.

PRISM — Pull Request Insight & Supervision Machine | *FastAPI · React · TypeScript · GitHub Actions · Gemini*

- Built a GitHub-integrated full-stack code review system combining a React frontend with a modular FastAPI backend, surfacing AI-generated risk classifications and plain-English PR summaries to developers.
- Architected a backend separating prompt execution, scoring, and synthesis layers to support evolving review heuristics and clean frontend consumption via structured JSON responses.
- Integrated GitHub Actions into the CI pipeline for automated review triggers, with unit-level testing and a scoring rubric to validate output consistency before surfacing results.

ERoute — Google ‘Build with AI’ Winner, Hack Canada 2026 | *Next.js · TypeScript · Mapbox GL · MongoDB · LLM APIs*

- Shipped a full-stack emergency routing platform under hackathon constraints using Next.js/TypeScript, integrating Mapbox GL for real-time geospatial visualization of hospital congestion and demand heatmaps.
- Built a voice-based LLM triage pipeline with a MongoDB backend, implementing structured data flow from patient input through urgency classification to dynamic hospital routing displayed on the frontend dashboard.

EDUCATION

Bachelor of Science, Computer Science

University of Toronto

Sept 2025 – Present

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, HTML, CSS
- **Frameworks:** React, Next.js, FastAPI, Flask, Tailwind CSS
- **Full-Stack:** REST API design, component-driven UI, responsive layouts, state management, CI/CD, GitHub Actions
- **Tools:** Git, GitHub, Vercel, Railway, MongoDB, VS Code, Notion
- **AI / ML:** XGBoost, LLM API integration (Gemini, Claude, OpenAI), prompt engineering, RAG, hallucination detection